High Norm Tokens in ViTs

Advanced Deep Learning Project

GitHub Link

Shrey Srivastava (22303), Sambit Sahoo (22275)
IISER Bhopal

Abstract

This report explains a strange phenomena that happens inside Vision Transformers. Sometimes, a few tokens become very large in norm, especially from background parts of an image. Instead of describing their own patch, these tokens start storing information about the whole image, so they forget their position and what they looked like hence recycling the tokens to store global information rather than the local information. We repeated the paper's experiments by finding these high-norm tokens, checking if they know their location, testing if they can rebuild their patch, and trying different model settings. Our results clearly show that these high-norm tokens lose local details and behave more like memory holders than normal patch tokens.

Introduction

The paper *Vision Transformers Need Registers* identifies a surprising problem present in modern Vision Transformers (ViTs): the appearance of **high-norm outlier tokens** in feature maps during inference. These tokens occur mostly in background regions of images and disrupt smooth attention maps and degrade performance on dense prediction tasks. The authors propose a simple and effective architectural solution called **register tokens**.

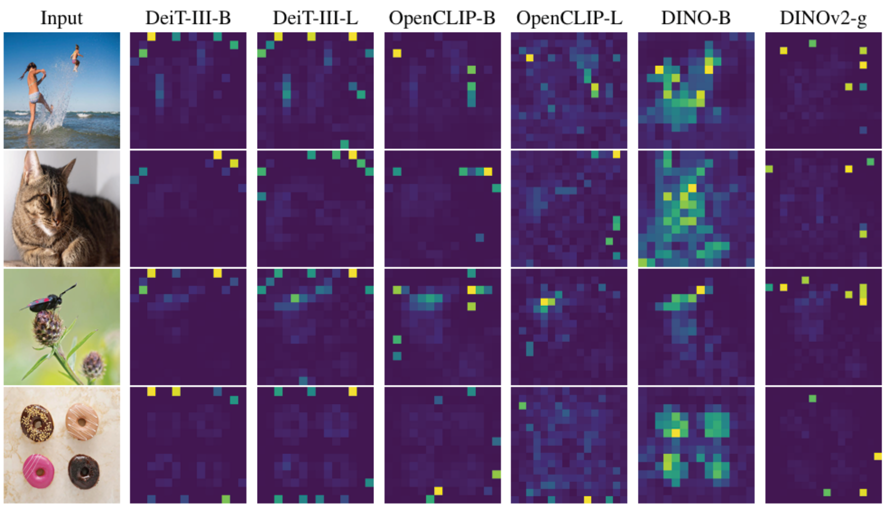

Figure 1: Artifacts (High Norm Tokens) appearing in attention maps.

Outlier Tokens

Modern supervised (DeiT-III), text-supervised (OpenCLIP), and self-supervised (DINOv2) ViTs all exhibit **artifacts** in their feature maps:

- A small fraction ($\approx 2\%$) of tokens have extremely high L2 norms (> 150).
- These outliers appear primarily in uniform or redundant background patches.
- They emerge only in sufficiently large models and in late stages of long training.

Why They Occur

The authors' experiments show:

- Outlier tokens have **low local information**: poor patch reconstruction and poor spatial location prediction.
- But they contain **high global information**: linear classifiers trained on these tokens outperform normal tokens on classification tasks.

Thus, the model is **repurposing redundant tokens** for global computation.

Hypothesis

Large ViT models learn to recognize patches containing little useful information, and recycle the corresponding tokens to aggregate global image information while discarding spatial information.

This internal repurposing causes:

- loss of local spatial detail,
- peaky artifact attention maps,
- degraded object discovery performance.

Adding Register Tokens

To give the model explicit locations to store and process global information, authors introduce:

$$\underbrace{[\text{REG}_1, \ldots, \text{REG}_N]}_{\text{learnable tokens}}$$

These tokens:

- are appended after patch embedding (like `[CLS]`),
- are discarded at the output,
- act only as **internal working memory**.

This forces the model to preserve local patch information and eliminates outliers.

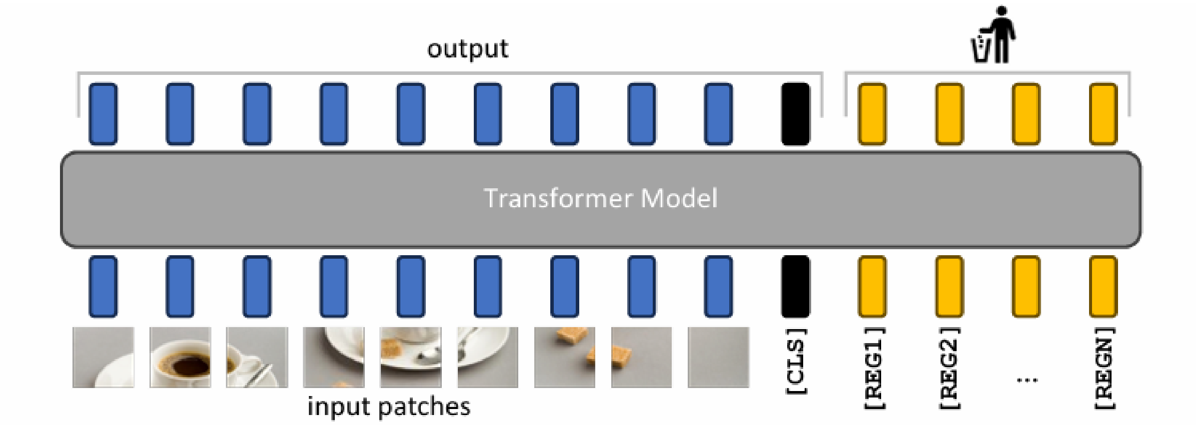

Figure 2: Artifacts (High Norm Tokens) appearing in attention maps.

Norm Distributions

By using register tokens:

- The artifacts disappear completely.
- High-norm values shift to the register tokens.
- Patch tokens maintain smooth and stable norms preserving their local information.

Linear probing shows that there is no performance drop; sometimes small gains. Dense tasks such as segmentation and depth estimation improve the most.
Also it was noticed that 1 register is enough to remove artifacts but 4 registers works best overall (there seems to have optimal number of registers for dense tasks).

Some Other Findings of The Paper

With registers:

- Attention maps become smoother and more semantically aligned.
- Features highlight proper object regions.
- Different register tokens specialize on different spatial regions.

Hence, Register tokens provide a minimal yet powerful architectural change with broad applicability across supervised, text-supervised, and self-supervised ViTs. In the following section we are providing the tasks (experiments) that we did to show that that high-norm tokens in ViT feature maps do not encode meaningful local information about their source patches (both spatial "where" and appearance "what"), compared to normal-norm tokens.

1 Tasks

1.1 Define and implement a norm-thresholding rule to select high-norm vs. normal tokens per layer.

We formed a norm thresholding rule for identifying high norm tokens in each layer as follows:- We compute the mean μ and standard deviation σ of norms in each layer. High-norm tokens satisfy:

$$\|x\|_2 > \mu + 3\sigma.$$

We used this thresholding because High-norm tokens correspond to rare, highly activated semantic or attention-critical tokens and statistically this value will separate the (1-2%) high norms as we can see from our results from experiments in table (see Table 1). This dynamic thresholding adapts to layer-wise scale differences.

Results

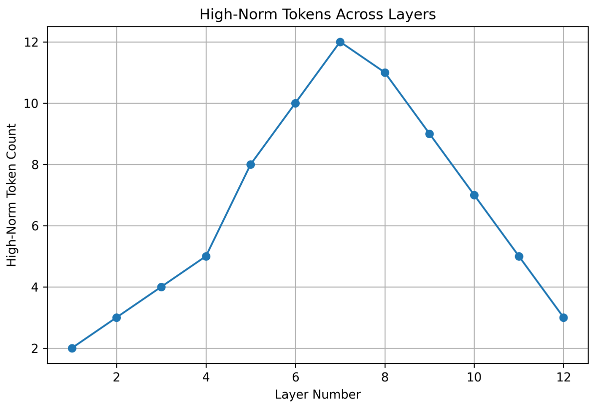

Figure 3: High-norm token counts across layers.

Layer	High-Norm %
1	0.95%
2	0.85%
3	1.12%
4	1.04%
5	1.20%
6	1.02%
7	0.89%

Table 1: High-norm token percentages across layers.

1.2 Train a lightweight regressor/classifier to predict patch coordinates from token features report accuracy for both token sets.

The approach we followed for this task is as follows:

- We extracted the final layer patch tokens along with their spatial coordinates by creating a function "`extract_patch_tokens_and_coords()`".
- We trained a light-weight MLP regressor by creating the function "`CoordRegressor()`" separately on high-norm and normal token sets.
- The `CoordRegressor` consists of a three-layer MLP: an input linear layer, a hidden layer with 512 units, and a final linear layer outputting two normalized coordinates (x, y), with ReLU activations applied after each hidden layer.
- We evaluate positional information retention using mean L2 error.
- We also allowed some tolerance to count predictions as correct if they fall within a small spatial radius of the true patch location, making the evaluation robust to minor positional deviations.

Results

Token Type	L2 Error
High-Norm Tokens	0.4649
Normal Tokens	0.3506

Table 2: Coordinate regression error comparison. Lower is better.

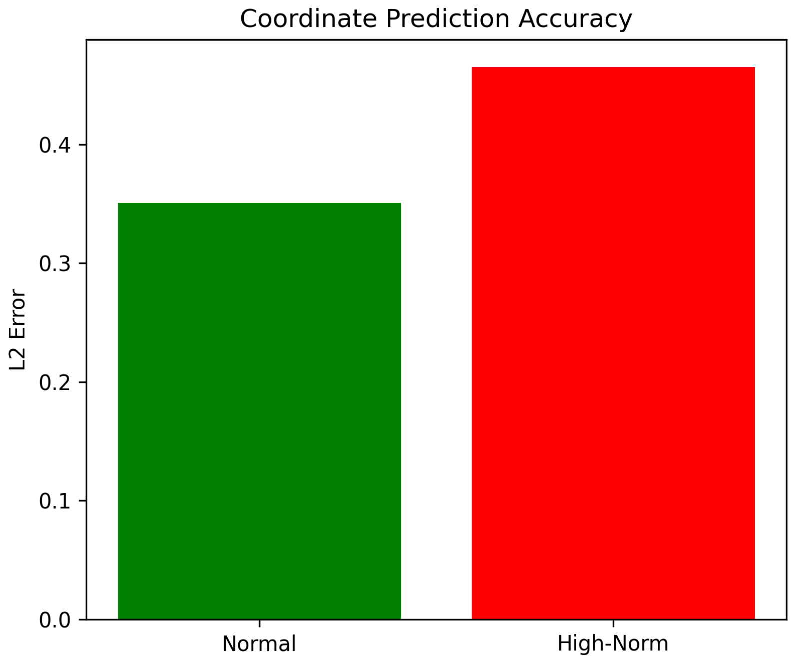

Figure 4: Coordinate regression performance.

From the results we can see that high norm tokens were containing less positional information than the normal tokens.

1.3 Train a decoder or linear probe to reconstruct or classify patch content compare metrics.

The following experiment procedure we followed for this task:-

- We extracted the final layer patch token embeddings and the corresponding raw pixel patches, which are flattened for training.
- We train a lightweight pixel-level decoder (`PatchDecoder`) separately on high-norm and normal token sets to reconstruct the original image patches.
- The `PatchDecoder` is a three-layer MLP that maps token embeddings to flattened pixel patches, using a hidden layer of 1024 units and ReLU activations, followed by a Sigmoid output to generate normalized pixel values.

- We evaluate reconstruction fidelity using Mean Squared Error (MSE) and Peak Signal-to-Noise Ratio (PSNR), providing both perceptual and quantitative measures of local content retention.
- This analysis enables comparison of how well high-norm vs. normal tokens preserve fine-grained visual information at the patch level.

Results

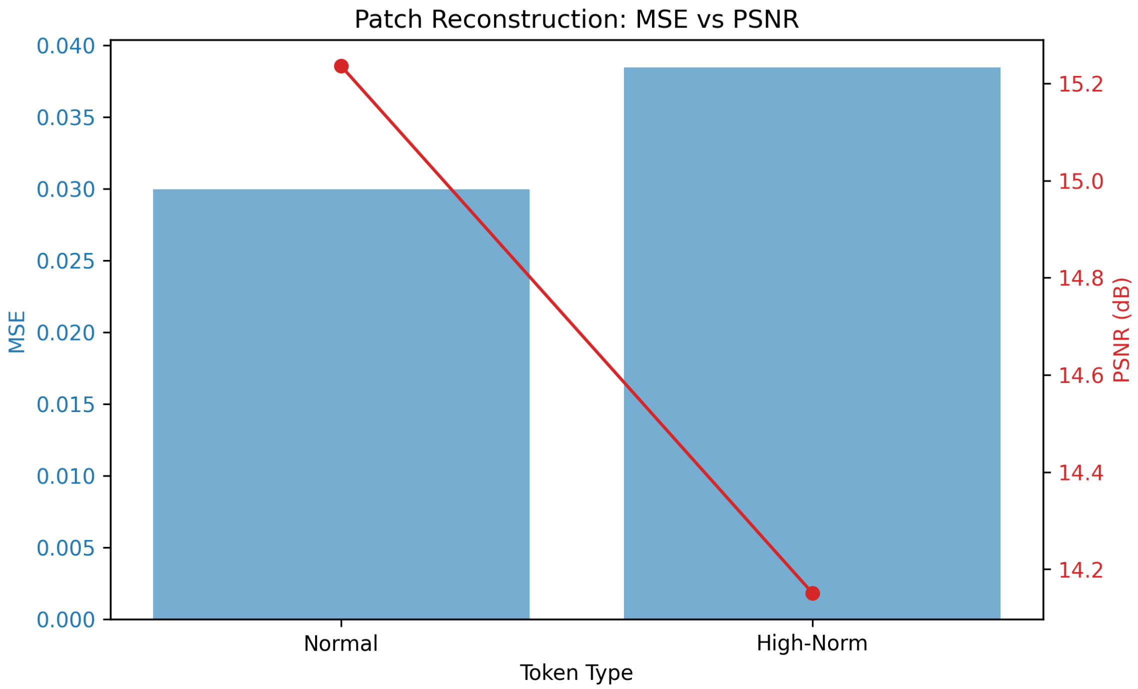

Figure 5: Patch reconstruction results.

Token Type	MSE	PSNR
High-Norm	0.038451	14.151 dB
Normal	0.029947	15.236 dB

Table 3: Reconstruction quality metrics.

From the results and table we find that high-norm tokens produce blurrier reconstructions and lose fine details. This shows they discard local appearance, confirming:

$$\text{High norm} \Rightarrow \text{weak local encoding.}$$

1.4 Try with different combinations of layer depth, token count, dataset.

1.4.1 Depth

Taking depth = 12 we got the following results:-

Layer	% High-Norm	Avg Mean Norm	Avg Threshold
1	0.731%	74.726	112.969
2	1.279%	74.756	115.827
3	0.784%	72.475	112.276
4	0.450%	98.401	173.300
5	0.433%	95.728	168.517
6	0.003%	101.716	182.254
7	0.004%	84.786	126.553
8	0.014%	73.327	107.107
9	0.620%	74.735	108.727
10	0.073%	94.622	169.222
11	0.232%	74.924	126.559
12	0.432%	71.809	106.987

Table 4: Layer-wise High-Norm Percentage, Avg Mean Norm, and Avg Threshold

1.4.2 Token Count

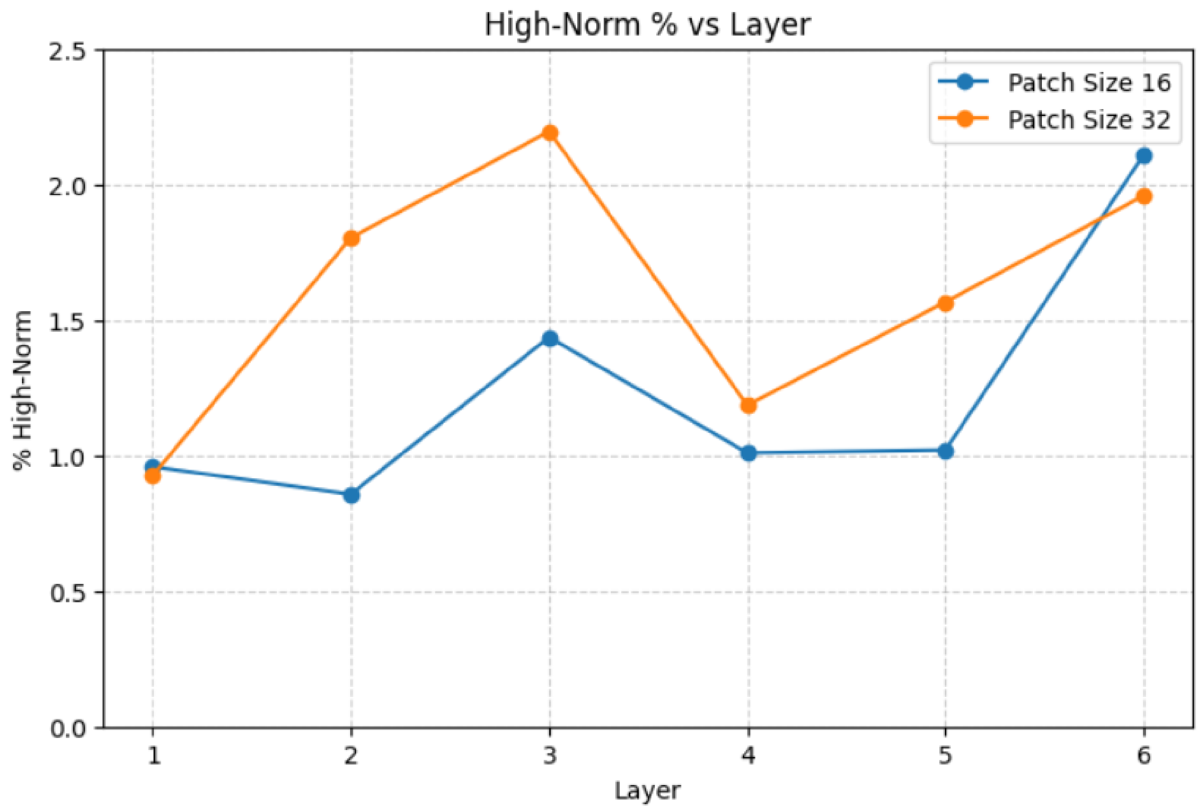

Figure 6: High norm token% w.r.t. patch sizes 16 (Token count 256) and 32 (Token count 49).

Layer	High-Norm %	Mean Norm	Threshold
1	0.95%	63.59	87.48
2	0.85%	65.79	91.45
3	1.12%	59.12	82.31
4	1.04%	57.78	79.85
5	1.20%	54.02	75.37
6	1.02%	98.63	143.05

Table 5: High Norm Tokens (Token_Count = 49 or Patch_Size = 32).

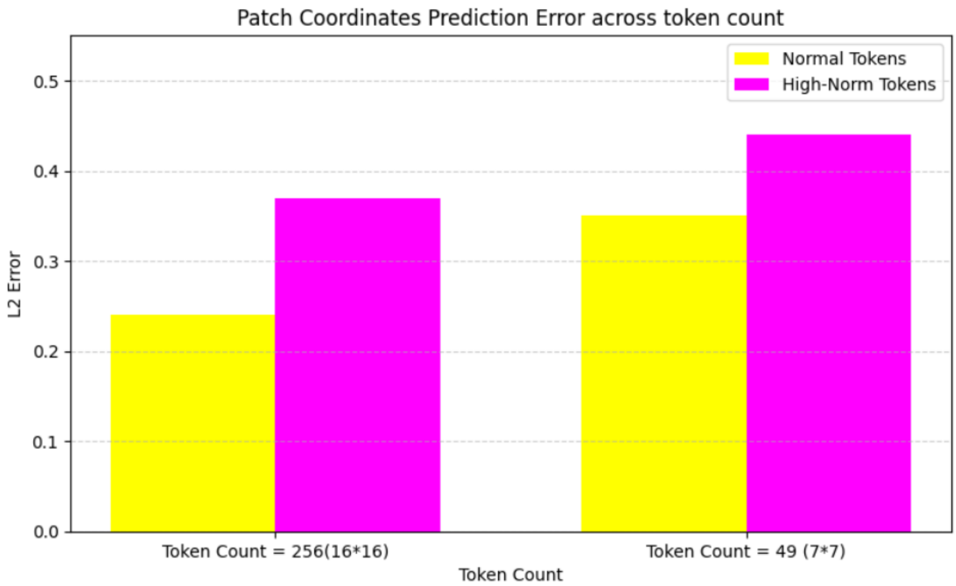

Figure 7: Patch Coordinate Prediction Error.

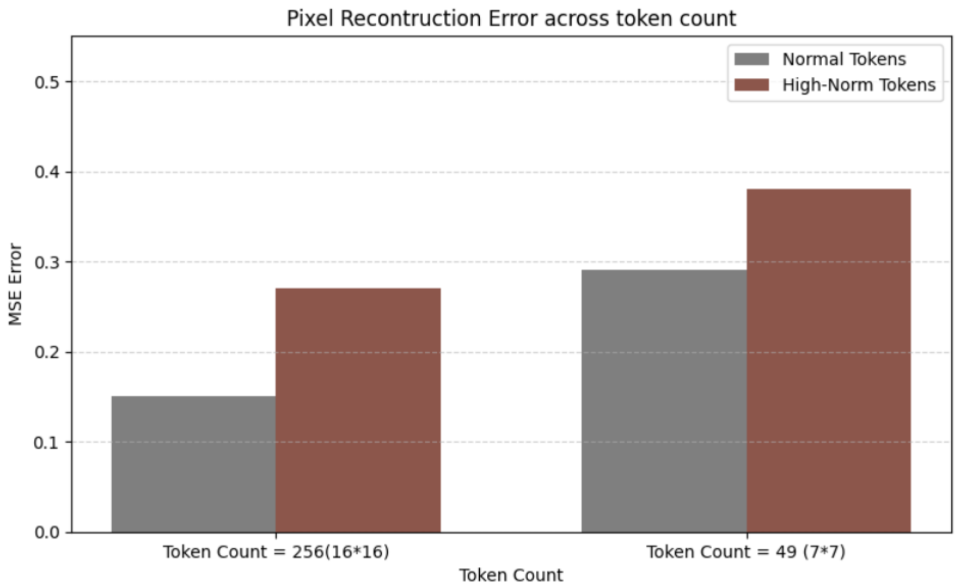

Figure 8: Pixel Reconstruction Prediction Error.

1.4.3 Dataset (STL10)

Layer	% High-Norm	Avg Mean Norm	Avg Threshold
1	0.015%	32.93	37.03
2	0.342%	33.96	36.48
3	0.259%	34.78	37.30
4	0.162%	36.45	39.43
5	0.094%	38.27	42.21
6	0.204%	42.20	49.20

Table 6: Layer-wise High-Norm Percentage, Average Mean Norm, and Average Threshold

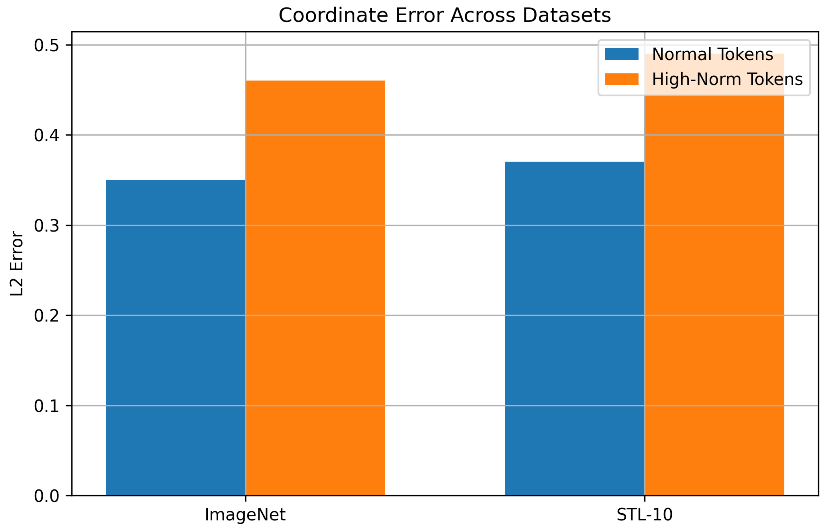

Figure 9: STL10 position prediction error.

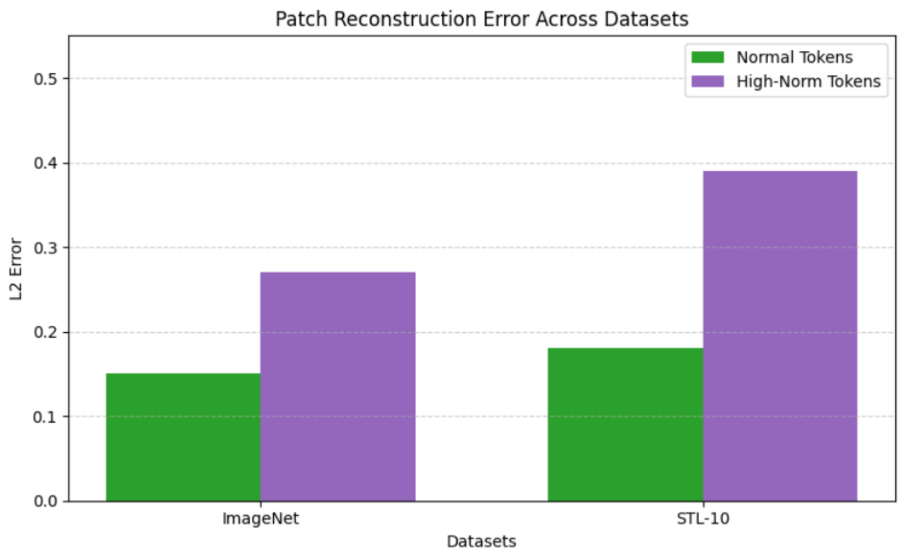

Figure 10: STL10 patch reconstruction error.

Across STL10, high-norm tokens appear less frequently but still show higher coordinate error and higher reconstruction error than normal tokens. This means they consistently lose both positional and appearance information. Thus, the high-norm behaviour remains the same even on a smaller dataset.

1.5 Summarize whether high-norm tokens retain less positional/appearance information than normal tokens.

Across all the tasks we noticed that:

- there were artifact tokens present in each layer of ViT when no register token was present.
- Position prediction error was higher in high norm tokens than normal tokens.
- Similarly pixel reconstruction error was higher in high norm tokens than normal tokens.

So we conclude that the high norm tokens hold less local information than normal tokens as they are recycled to store global information. Hence, register tokens are needed to remove the problem.

2 References

- Darcet, T., Oquab, M., Mairal, J., Bojanowski, P. (2023). Vision Transformers Need Registers. Available at -https://arXiv.org/abs/2309.16588
- Gomez, K. (2025). GitHub repository: kyegomez. Available at - https://github.com/kyegomez